

Individual assignment for Theoretical Modeling for Cognitive Science (BKI334)

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Word count: $n = 1062$ words.

1 Introduction

Categorization is a cognitive process that allows people to organize the infinite variety of the world (objects, phenomena, events) into meaningful categories for simplified perception, decision-making, memory, and learning. Understanding categorization would not only better understand the human brain and realize the consequences that follow from this, such as decision-making or memory, and be a step towards improving AI learning based on categorization, i.e., clustering.

The verbal theory underlying this work assumes that humans form categories spontaneously for two goals: to maximize similarity between objects within one category and to minimize similarity between objects from different categories. The formalization developed by Group 15 reflects this idea into a computational-level model but contains certain implicit errors, which will be addressed and clarified.

The main goal of this paper is a detailed formal analysis of the formalization "Categorization" of Group 15. The next stages of the work will be correction of possible inaccuracies, a formal analysis of this model, and a discussion of the implications for cognitive science.

2 Formal theory

The original model with detailed explanations and some minor fixes:

Input:

- A set of objects $O = \{o_1, o_2, \dots, o_n\}$, $o \in O$.

This set can contain any finite n number of elements representing abstract or real objects.

- Similarity function $s : O \times O \rightarrow [0, 1]$; $s(o, o')$ - similarity measure between objects o and o' .

This function takes a pair of objects and returns a numeric value, where 0 means complete dissimilarity, and 1 means maximum similarity.

Output:

- A set of categories $C \subsetneq \mathcal{P}(O) \setminus \{\emptyset\}$.

$C = \{c_1, c_2, \dots, c_k\}$ where each $c \in C$ is a subset of objects, such that:

Each category can be viewed as a non-empty subset of a set of objects. Categories must satisfy four following conditions:

$$1. W(C) = \operatorname{argmax}_C \sum_{c \in C} \sum_{\substack{o, o' \in c \\ o \neq o'}} s(o, o')$$

Maximizes the sum of similarities between objects within the same category should be as similar to each other as possible. This is expressed by maximizing the sum of the similarities between objects within the same category. Corrected version excludes self-similarity comparison.

$$2. B(C) = \operatorname{argmin}_{C'} \sum_{\substack{c, c' \in C' \\ c \neq c'}} \sum_{o \in c} \sum_{o' \in c'} s(o, o')$$

Objects from different categories should be as dissimilar to each other as possible. This is expressed by minimizing the sum of similarities between objects from different categories.

$$3. W_{\text{norm}}(C) = \frac{1}{N_W} W(C), \text{ where } N_W = \sum_{c \in C} \binom{|c|}{2} \text{ the largest number of unique pairs within the categories.}$$

$$B_{\text{norm}}(C) = \frac{1}{N_B} B(C), \text{ where } N_B = \sum_{\substack{c, c' \in C \\ c \neq c'}} |c| \cdot |c'| \text{ the total number of between-category pairs.}$$

Both normalized functions return values in the range $[0, 1]$, which allows to compare different configurations of categories.

$$4. \forall c \neq c' \in C : c \cap c' = \emptyset$$

No object belongs to more than one category (categories are mutually exclusive). This is expressed by empty intersection of all categories.

Before $\bigcap_{c \in C} c = \emptyset$ was used, but it did not prohibit the object from belonging to many categories.

$$5. \bigcup_{c \in C} c = O$$

Every object in O is assigned to exactly one category. This is expressed by combining all categories, which should give the whole set of objects.

Then the optimal partition C^* is defined as:

$$C^* = \operatorname{argmax}_{C \subset \mathcal{P}(O) \setminus \{\emptyset\}} F_{\text{norm}}(C) = W_{\text{norm}}(C) - \lambda \cdot B_{\text{norm}}(C)$$

where $\lambda \geq 0$ is a hyperparameter that regulates the balance between $W_{\text{norm}}(C)$ and $B_{\text{norm}}(C)$.

This formalization describes categorization as an optimization problem: one must find a partition of the set of objects that simultaneously strengthens $W_{\text{norm}}(C)$ and weakens $B_{\text{norm}}(C)$.

3 Formal analysis

3.1 Hypothesis

1. $H0$: The optimal categorization C^* as a function of the hyperparameter λ is monotonic, with an increase in λ (i.e., increase in penalty for $B_{\text{norm}}(C)$), the average between-category similarity $B_{\text{norm}}(C)$ does not increase (monotonously decreases or remains constant).
2. $H1$: If $\lambda = +\infty$, optimal division will be non-trivial.

3.2 Proof $H1$: increase in λ monotonously decreases $B_{\text{norm}}(C)$

For this proof, Python code was developed for conducting simulations. Each simulation had three objects A, B, C and all five partitions were listed.

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All possible partitions for objects ['A', 'B', 'C']:
1. P1: {A},{B},{C}
2. P2: {A,B},{C}
3. P3: {A,C},{B}
4. P4: {B,C},{A}
5. P5: {A,B,C}
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Figure 1: Five possible partitions for objects A, B, C.

For simplicity only three partitions will be analyzed (for partition where two categories - does not matter which one of three). The Figure 2 gives an illustration of three partitions.

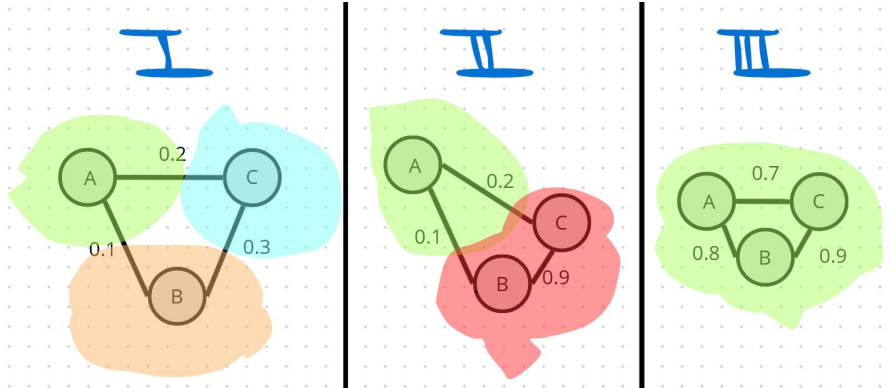


Figure 2: Three example partitions (I, II, III) with different category numbers (shown by colors) and manually assigned similarity values (shown near edges).

The next part of code was the encoded mathematics of formalization. Then code outputs tables for each λ , where was five rows of all partitions and each column is a value for each of $W_{\text{norm}}(C)$, $B_{\text{norm}}(C)$, $F_{\text{norm}}(C)$, *numberofcategories*. Below each table, optimal partition for corresponding λ was chosen by maximum and showed. The Figure 3 gives some examples of that.

TABLE FOR $\lambda = 0$

Partition	Within_norm	Between_norm	F_norm(C)	Categories
{A},{B},{C}	0.000	0.200	0.000	3
{A,B},{C}	0.100	0.250	0.100	2
{A,C},{B}	0.200	0.200	0.200	2
{B,C},{A}	0.300	0.150	0.300	2
{A,B,C}	0.200	0.000	0.200	1

Optimal partition for $\lambda = 0$:
{B,C},{A}
F_norm(C) = 0.300
Within_norm = 0.300, Between_norm = 0.150
Category number: 2

TABLE FOR $\lambda = 1$

Partition	Within_norm	Between_norm	F_norm(C)	Categories
{A},{B},{C}	0.000	0.200	-0.200	3
{A,B},{C}	0.100	0.250	-0.150	2
{A,C},{B}	0.200	0.200	0.000	2
{B,C},{A}	0.300	0.150	0.150	2
{A,B,C}	0.200	0.000	0.200	1

Optimal partition for $\lambda = 1$:
{A,B,C}
F_norm(C) = 0.200
Within_norm = 0.200, Between_norm = 0.000
Category number: 1

TABLE FOR $\lambda = 0$

Partition	Within_norm	Between_norm	F_norm(C)	Categories
{A},{B},{C}	0.000	0.400	0.000	3
{A,B},{C}	0.100	0.550	0.100	2
{A,C},{B}	0.200	0.500	0.200	2
{B,C},{A}	0.900	0.150	0.900	2
{A,B,C}	0.400	0.000	0.400	1

Optimal partition for $\lambda = 0$:
{B,C},{A}
F_norm(C) = 0.900
Within_norm = 0.900, Between_norm = 0.150
Category number: 2

TABLE FOR $\lambda = 1$

Partition	Within_norm	Between_norm	F_norm(C)	Categories
{A},{B},{C}	0.000	0.400	-0.400	3
{A,B},{C}	0.100	0.550	-0.450	2
{A,C},{B}	0.200	0.500	-0.300	2
{B,C},{A}	0.900	0.150	0.750	2
{A,B,C}	0.400	0.000	0.400	1

Optimal partition for $\lambda = 1$:
{B,C},{A}
F_norm(C) = 0.750
Within_norm = 0.900, Between_norm = 0.150
Category number: 2

TABLE FOR $\lambda = 0$

Partition	Within_norm	Between_norm	F_norm(C)	Categories
{A},{B},{C}	0.000	0.800	0.000	3
{A,B},{C}	0.800	0.800	0.800	2
{A,C},{B}	0.700	0.850	0.700	2
{B,C},{A}	0.900	0.750	0.900	2
{A,B,C}	0.800	0.000	0.800	1

Optimal partition for $\lambda = 0$:
{B,C},{A}
F_norm(C) = 0.900
Within_norm = 0.900, Between_norm = 0.750
Category number: 2

TABLE FOR $\lambda = 1$

Partition	Within_norm	Between_norm	F_norm(C)	Categories
{A},{B},{C}	0.000	0.800	-0.800	3
{A,B},{C}	0.800	0.800	0.000	2
{A,C},{B}	0.700	0.850	-0.150	2
{B,C},{A}	0.900	0.750	0.150	2
{A,B,C}	0.800	0.000	0.800	1

Optimal partition for $\lambda = 1$:
{A,B,C}
F_norm(C) = 0.800
Within_norm = 0.800, Between_norm = 0.000
Category number: 1

Figure 3: Tables for all partitions(I, II, III) for $\lambda = 0$ and $\lambda = 1$.

The final summary table contains Optimal Partition for each λ values and same columns with value information for each of formalizations components. λ is ordered ascending manually, while descending order of $B_{\text{norm}}(C)$ proofs $H0$. This also can be seen on the graphs, where can be seen that with bigger λ values, $B_{\text{norm}}(C)$ decreases. Figure 4 shows final output for each scenario and Figure 5 graphs of those.

I	SUMMARY TABLE: OPTIMAL PARTITIONS					
	λ	Optimal Partition	Categories	Within_norm	Between_norm	F_norm(C)
	0.0	{B,C},{A}	2	0.300	0.150	0.300
	0.5	{B,C},{A}	2	0.300	0.150	0.225
	1.0	{A,B,C}	1	0.200	0.000	0.200
	2.0	{A,B,C}	1	0.200	0.000	0.200
	4.0	{A,B,C}	1	0.200	0.000	0.200
	6.0	{A,B,C}	1	0.200	0.000	0.200
	8.0	{A,B,C}	1	0.200	0.000	0.200
	10.0	{A,B,C}	1	0.200	0.000	0.200
II	SUMMARY TABLE: OPTIMAL PARTITIONS					
	λ	Optimal Partition	Categories	Within_norm	Between_norm	F_norm(C)
	0.0	{B,C},{A}	2	0.900	0.150	0.900
	0.5	{B,C},{A}	2	0.900	0.150	0.825
	1.0	{B,C},{A}	2	0.900	0.150	0.750
	2.0	{B,C},{A}	2	0.900	0.150	0.600
	4.0	{A,B,C}	1	0.400	0.000	0.400
	6.0	{A,B,C}	1	0.400	0.000	0.400
	8.0	{A,B,C}	1	0.400	0.000	0.400
	10.0	{A,B,C}	1	0.400	0.000	0.400
III	SUMMARY TABLE: OPTIMAL PARTITIONS					
	λ	Optimal Partition	Categories	Within_norm	Between_norm	F_norm(C)
	0.0	{B,C},{A}	2	0.900	0.750	0.900
	0.5	{A,B,C}	1	0.800	0.000	0.800
	1.0	{A,B,C}	1	0.800	0.000	0.800
	2.0	{A,B,C}	1	0.800	0.000	0.800
	4.0	{A,B,C}	1	0.800	0.000	0.800
	6.0	{A,B,C}	1	0.800	0.000	0.800
	8.0	{A,B,C}	1	0.800	0.000	0.800
	10.0	{A,B,C}	1	0.800	0.000	0.800

Figure 4: Optimal portioning tables for three analyzed scenarios.

3.3 Proof $H1 : \lambda = +\infty$

$$C^* = \lim_{\lambda \rightarrow +\infty} F_{\text{norm}}(C) = \lim_{\lambda \rightarrow +\infty} [W_{\text{norm}}(C) - \lambda \cdot B_{\text{norm}}(C)].$$

Simplify this to:

$$C^* = \arg \max_{C \in \mathcal{P}(O) \setminus \{\emptyset\}} F_{\text{norm}}(C) = W_{\text{norm}}(C) - \lambda \cdot B_{\text{norm}}(C) = W_{\text{norm}}(C) - \infty \cdot B_{\text{norm}}(C) \approx -\infty \cdot B_{\text{norm}}(C)$$

For $\lambda \rightarrow +\infty$, the term $-\lambda \cdot B_{\text{norm}}(C)$ dominates. To maximize $F_{\text{norm}}(C)$, it is necessary to minimize $B_{\text{norm}}(C)$, because of huge factor and subtraction from $W_{\text{norm}}(C)$. $B_{\text{norm}}(C) = 0$ if and only if all objects are in one category and there are no other category to compare with.

With $\lambda = +\infty$ model will create one large category containig all of the objects, this proofs $H1$.

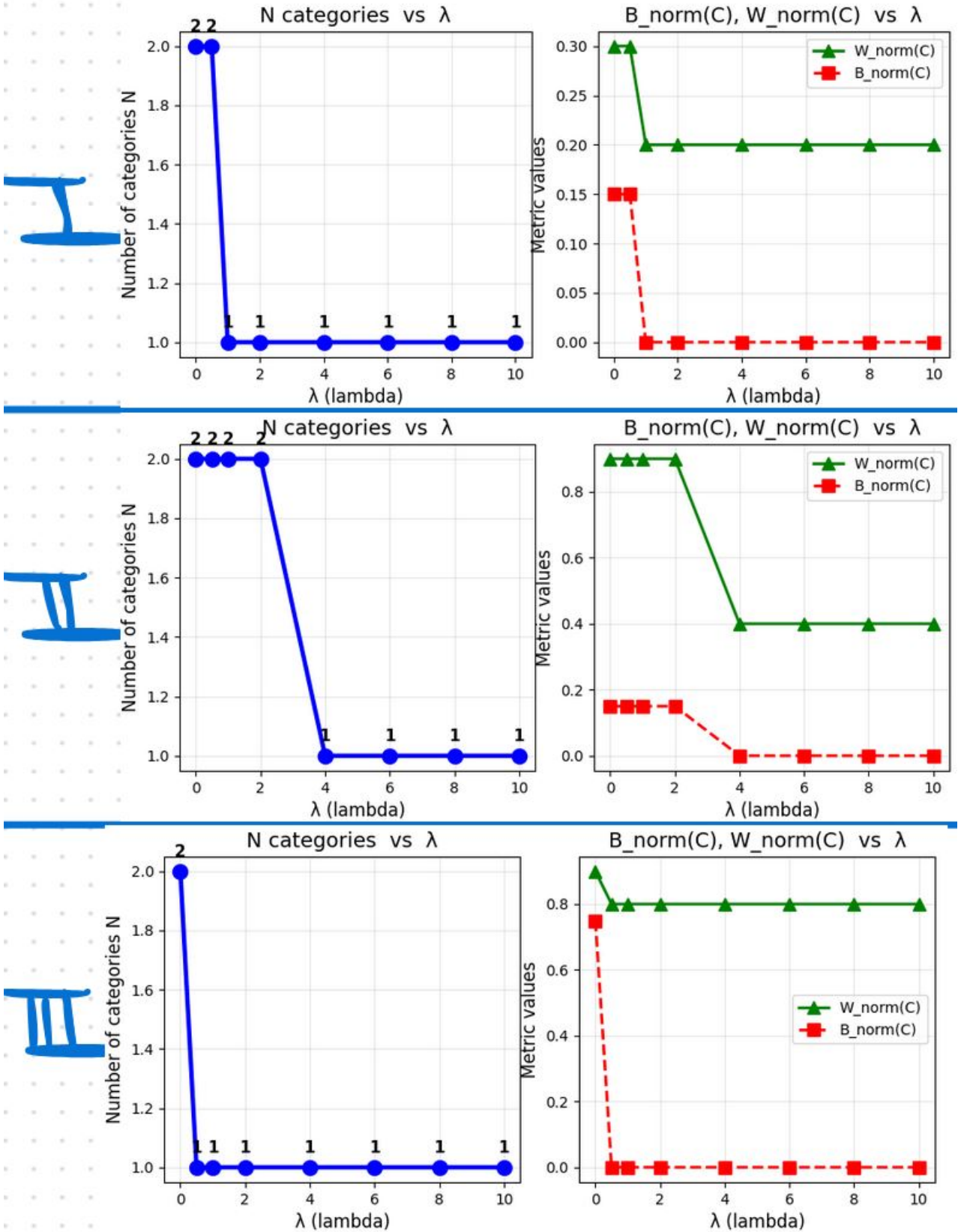


Figure 5: Left graph shows dependence between N number of categories and λ . Right graph shows dependence between $W_{\text{norm}}(C)$, $B_{\text{norm}}(C)$ and λ . Each row correspond one scenario.

4 Discussion

Analysis confirms hypotheses $H0$ and $H1$. Hypotheses provide strong evidence of the function's monotonicity: as λ increases, the "penalty" for $B_{\text{norm}}(C)$ grows, and the model strives to

minimize it. At $\lambda \rightarrow +\infty$, $B_{\text{norm}}(C)$ becomes zero, merging all objects into a single category, which is a non-trivial solution.

λ acts as resolution parameter, aligning with cognitive theories of categorization. Furthermore, the graphs show that transition to one category can occur at different λ values for different scenarios, indicating the potential for investigating critical λ thresholds when only one category produces. Second graph proves decreasing trend for B_{norm} .

The model's behavior strictly follows from its mathematical formulation. It reflects the inherent trade-off of two goals in categorization. The model illustrates how a cognitive system can abruptly shift strategies when the importance (λ) of one principle changes.

Main take-home message: The formalization by Group 15, after corrections, is internally consistent and theoretically sound. The model adequately translates the verbal theory into a mathematical framework and demonstrates predictable, intuitive behavior. The simulation results confirm that the parameter λ governs the balance between categorization goals, and the limiting behavior at $\lambda \rightarrow +\infty$, leading to a trivial single category, is a strict and logical consequence of the formalization itself.